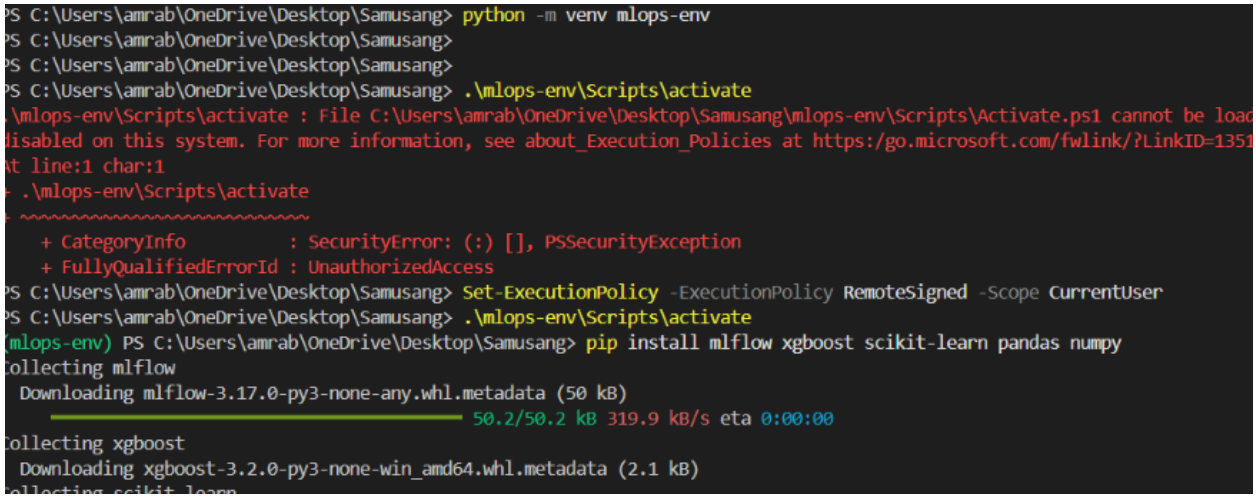
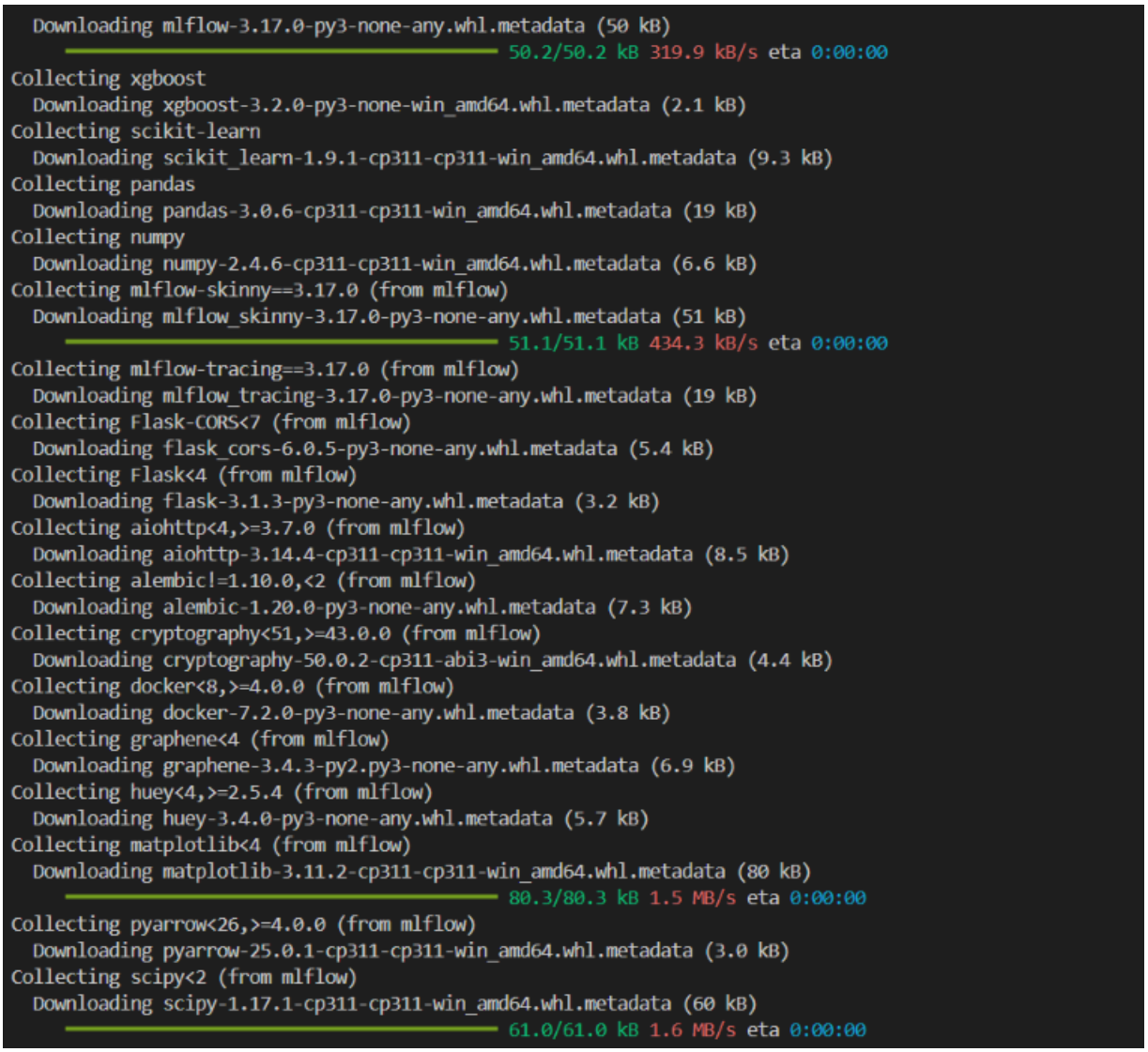
Step 1: Initialize the Environment





Step 2: Start the MLflow Tracking Server

```
(mlops-env) PS C:\Users\amrab\OneDrive\Desktop\Samusang> mlflow server --backend-store-uri sqlite:///mlruns.db --default-artifact-root ./artifacts --host 127.0.0
.1 --port 5000
Registry store URI not provided. Using backend store URI.
2026/10/10 01:08:03 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/10/10 01:08:03 INFO mlflow.store.db.utils: Updating database tables
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-ho
sts and --cors-allowed-origins.
2026/10/10 01:08:12 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start wit
hout job execution support. Errors will be surfaced at job invocation time.
2026/10/10 01:08:12 INFO:     Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/10 01:08:12 INFO:     Started parent process [32408]
C:\Users\amrab\OneDrive\Desktop\Samusang\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is d
eprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\amrab\OneDrive\Desktop\Samusang\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is d
eprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\amrab\OneDrive\Desktop\Samusang\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is d
eprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\amrab\OneDrive\Desktop\Samusang\mlops-env\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is d
eprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/10 01:08:28 INFO:     Started server process [26072]
2026/10/10 01:08:28 INFO:     Started server process [36528]
2026/10/10 01:08:28 INFO:     Waiting for application startup.
2026/10/10 01:08:28 INFO:     Waiting for application startup.
2026/10/10 01:08:28 INFO:     Started server process [11096]
2026/10/10 01:08:28 INFO:     Waiting for application startup.
2026/10/10 01:08:29 INFO:     Started server process [9016]
```

Step 3: The Training Script (train.py)

```
Problems  Output  Debug Console  Terminal  Ports
ktop\Samusang\mlops-env\Scripts\Activate.ps1)
(mlops-env) PS C:\Users\amrab\OneDrive\Desktop\Samusang> .\mlops-env\Scripts\python.exe train.py
2026/10/10 01:35:24 INFO mlflow.tracking.fluent: Experiment with name 'California_Housing_Optimization' does not exist. Creating a new experimen
t.
Fetching and splitting California Housing data...
2026/10/10 01:35:28 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
Run 1 Complete | Val RMSE: 0.5433
🏃 View run Run 1 at: http://127.0.0.1:5000/#/experiments/1/runs/71e31eeb81b64138a3f63d17c3cec972
🧪 View experiment at: http://127.0.0.1:5000/#/experiments/1
2026/10/10 01:35:37 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
Run 2 Complete | Val RMSE: 0.5215
🏃 View run Run 2 at: http://127.0.0.1:5000/#/experiments/1/runs/e72f1de3f2b54ebeb470ae88f3267861
🧪 View experiment at: http://127.0.0.1:5000/#/experiments/1
2026/10/10 01:35:44 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
Run 3 Complete | Val RMSE: 0.7006
🏃 View run Run 3 at: http://127.0.0.1:5000/#/experiments/1/runs/f79b291039cd49d9ba0fb5ed73cedaa9
🧪 View experiment at: http://127.0.0.1:5000/#/experiments/1
(mlops-env) PS C:\Users\amrab\OneDrive\Desktop\Samusang>
```

Step 4: Comparing and Promoting in the UI

Comparing 3 Runs from 1 Experiment

Visualizations

Parallel Coordinates Plot | Scatter Plot | Box Plot | Contour Plot

Parameters: learning_rate × max_depth ×

Metrics: val_rmse ×

Clear All



Run details

Comparing 3 Runs from 1 Experiment

Visualizations

Run details

Run ID:	71e31eeb81b64138a3f63d17c3cec972	e72f1de3f2b54ebeb470ae88f3267861	f79b291039cd49d9ba0fb5ed73cedaa9
Run Name:	Run 1	Run 2	Run 3
Start Time:	10/10/2026, 01:35:27 AM	10/10/2026, 01:35:37 AM	10/10/2026, 01:35:44 AM
End Time:	10/10/2026, 01:35:37 AM	10/10/2026, 01:35:44 AM	10/10/2026, 01:35:50 AM
Duration:	9.4s	7.1s	6.7s

Parameters

Show diff only

eval_metric	rmse	rmse	rmse
learning_rate	0.1	0.05	0.01
max_depth	3	5	7
objective	reg:squarederror	reg:squarederror	reg:squarederror

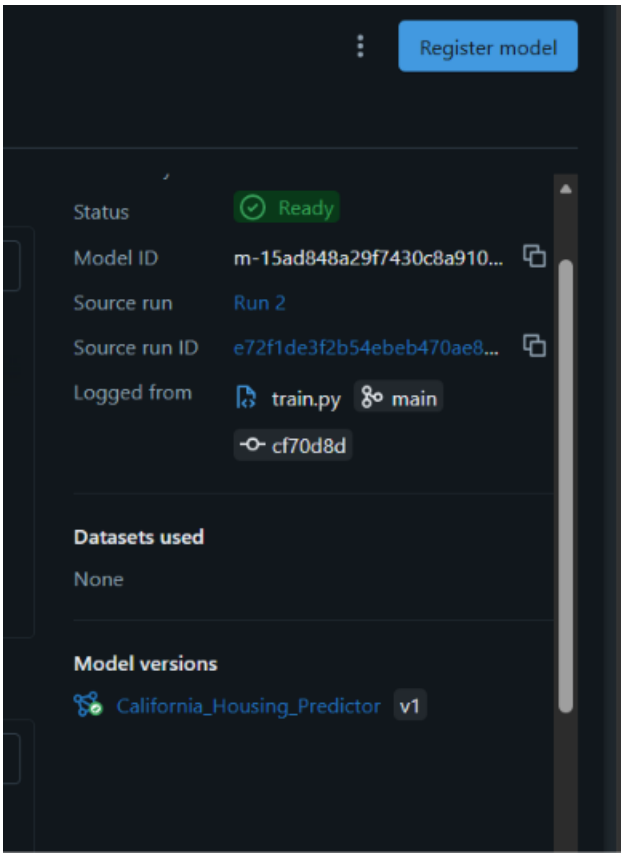
Metrics

Show diff only

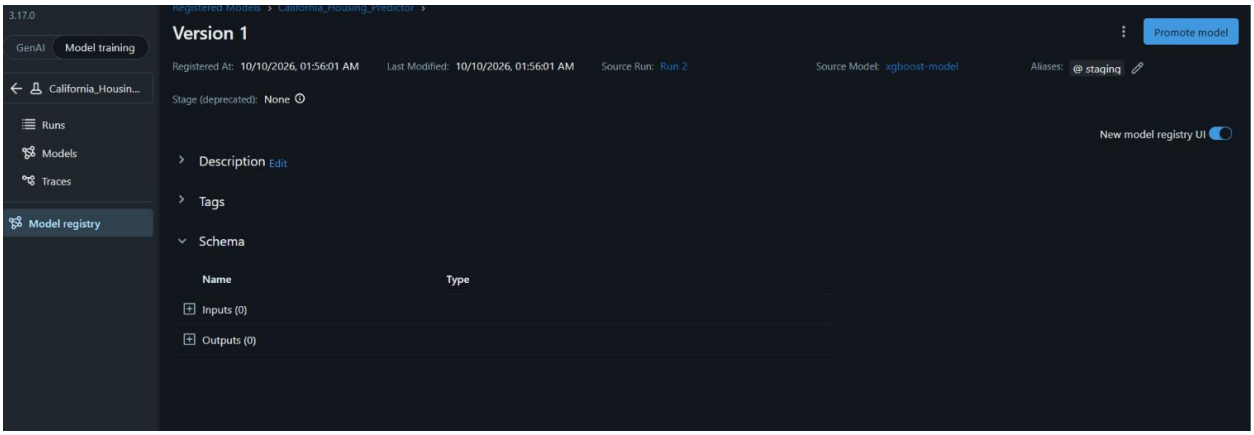
val_rmse	0.543	0.522	0.701



Create new model registry



4. Go to the Models tab, select Version 2, and register the model , then promote it and add the alias "staging" to the model.



Install minikube on windows if it’s not there

```
PS C:\WINDOWS\system32> cd C:\Users\amrab\OneDrive\Desktop\Samusang
PS C:\Users\amrab\OneDrive\Desktop\Samusang> minikube image build -t california-housing-api:v1 .
#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 387B 0.0s done
#1 DONE 0.1s

#2 [internal] load metadata for docker.io/library/python:3.10-slim
#2 DONE 24.2s

#3 [internal] load .dockerignore
#3 transferring context:
#3 transferring context: 2B done
#3 DONE 0.3s

#4 [internal] load build context
#4 transferring context: 279.98kB 0.0s done
#4 DONE 5.1s

#5 [1/7] FROM docker.io/library/python:3.10-slim@sha256:6ff506466f8b1e981719b468cc0023b8ef000b983cdec0eafe119d01f379e31d
#5 resolve docker.io/library/python:3.10-slim@sha256:6ff506466f8b1e981719b468cc0023b8ef000b983cdec0eafe119d01f379e31d 0.4s done
#5 DONE 5.3s

#5 [1/7] FROM docker.io/library/python:3.10-slim@sha256:6ff506466f8b1e981719b468cc0023b8ef000b983cdec0eafe119d01f379e31d
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 0B / 13.92MB 0.2s
#5 sha256:cc1cbe171e30bea48715fd96ecf842ad9fae425ce5a18917bee871684eb836f1 0B / 249B 0.2s
#5 sha256:1189ef607cc64ff2e2a214c12e5f46402dc316a1a62cc5adb7071616882df7ee 0B / 1.29MB 0.2s
#5 sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 0B / 29.84MB 0.2s
#5 sha256:cc1cbe171e30bea48715fd96ecf842ad9fae425ce5a18917bee871684eb836f1 249B / 249B 0.6s done
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 1.05MB / 13.92MB 2.9s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 2.10MB / 13.92MB 3.5s
#5 sha256:1189ef607cc64ff2e2a214c12e5f46402dc316a1a62cc5adb7071616882df7ee 1.05MB / 1.29MB 3.5s
#5 sha256:1189ef607cc64ff2e2a214c12e5f46402dc316a1a62cc5adb7071616882df7ee 1.29MB / 1.29MB 3.9s done
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 3.15MB / 13.92MB 5.0s
#5 sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 1.05MB / 29.84MB 5.3s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 4.19MB / 13.92MB 6.0s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 5.24MB / 13.92MB 7.7s
#5 sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 3.15MB / 29.84MB 8.0s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 6.29MB / 13.92MB 8.9s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 7.34MB / 13.92MB 10.2s
#5 sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 5.24MB / 29.84MB 10.6s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 8.39MB / 13.92MB 12.0s
#5 sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 7.34MB / 29.84MB 13.2s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 9.44MB / 13.92MB 13.6s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 10.49MB / 13.92MB 15.6s
#5 sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 9.44MB / 29.84MB 15.4s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 11.53MB / 13.92MB 17.5s
#5 sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 11.53MB / 29.84MB 17.8s
#5 sha256:79f4247a2db52342f23c9ff6911633949443ca83b454d185a05ed6fc8689ffca 12.58MB / 13.92MB 18.9s
```

kubectl get pods # 4. Check pod status (wait for 1/1 READY) kubectl get pods

```
ervice/housing-api-service created
S C:\Users\amrab\OneDrive\Desktop\Samusang> kubectl get pods
AME                                      READY   STATUS              RESTARTS   AGE
ousing-api-deployment-86c95c547d-cl69t   0/1     ContainerCreating   0          19s
ousing-api-deployment-86c95c547d-ksbw2   0/1     ContainerCreating   0          19s
S C:\Users\amrab\OneDrive\Desktop\Samusang> minikube service housing-api-service
```

Step 7: Testing the Deployment v



California Housing API 0.1.0 OAS 3.1

/openapi.json

default

GET /health Health Check

POST /predict Predict

Schemas

HTTPValidationError › Expand all object

HousingInput › Expand all object

ValidationError › Expand all object



GET /health Health Check

Parameters Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:57774/health' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:57774/health
```

Server response

Code	Details
200	Response body

```
{
  "status": "healthy",
  "model_loaded": true
}
```

Download

Response headers

```
content-length: 40
content-type: application/json
date: Fri,09 Oct 2026 23:44:12 GMT
server: uvicorn
```

Request URL

```
http://127.0.0.1:57774/predict
```

Server response

Code	Details
200	Response body

```
{
  "prediction": "$-26,810.98"
}
```

Download

Response headers

```
content-length: 28
content-type: application/json
date: Fri,09 Oct 2026 23:44:57 GMT
server: uvicorn
```

Responses

Code	Description	Links